Comparing Risk Models for GPU Kernel Replacement: Balanced Evidence Across Operating Budgets

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Abstract

GPU kernel replacement is commonly screened by comparing kernel outputs against a numerical tolerance, but local agreement does not ensure task-level agreement. We fit twelve risk models on 6,928 grouped perturbation observations and transfer their predictions to 2,739 real replacements. Scores use shared balanced replacement draws across 21 risk-budget pairs. Mean F1 ranges from 0.4867 for the prevalence baseline to 0.8484 for weighted k-nearest neighbors (KNN); the grouped multilayer perceptron (MLP) and multi-output random forest score 0.8460 and 0.8411. Rankings vary across budgets and metrics, while the leading mean F1 scores are close relative to within-budget repeat variation. The study characterizes screening trade-offs across model families without identifying a universally best model.

1 Introduction

Modern model-serving systems rely on highly optimized GPU kernels. Replacing a reference implementation with a faster kernel can improve throughput, but changes in reduction order, fused arithmetic, or intermediate precision can create floating-point discrepancies. Numerical analysis explains how such errors accumulate through computation graphs [1, 2, 3]. Recent inference studies also report output changes under different execution settings and low-precision arithmetic [4, 5].

Kernel-generation benchmarks make local correctness checks practical by comparing candidate-kernel outputs with a reference under a numerical tolerance [6, 7]. A passing check establishes that the compared tensors are close on the probe inputs. It does not establish that a replacement preserves the model's final class, answer, transcript, or correctness decision. The same numerical discrepancy can have different task-level effects depending on the affected operator and inference context.

We test whether risk models trained on controlled perturbations transfer to real replacements. The first endpoint is an *output change*: the task-level discrete output differs from the reference. The second is a *negative flip*: a request that the reference execution answers correctly becomes incorrect under the replacement. A negative flip is a subset of output changes and also depends on reference correctness. We compare the estimated rates of these outcomes because they capture different consequences of the same numerical discrepancy.

We use the supplied Kernel-Correctness study's 13-category operator-role taxonomy and balanced replacement protocol [8]. Each model maps maximum local discrepancy, task scope, dtype, and operator role to estimated output-change and negative-flip rates. We compare twelve fixed models spanning prevalence, monotone regression, linear and spline models, kernel regression, nearest neighbors, a neural network, single trees, and tree ensembles. Models are frozen before replacement outcomes are scored, and all are evaluated on the same balanced replacement draws. The compar-

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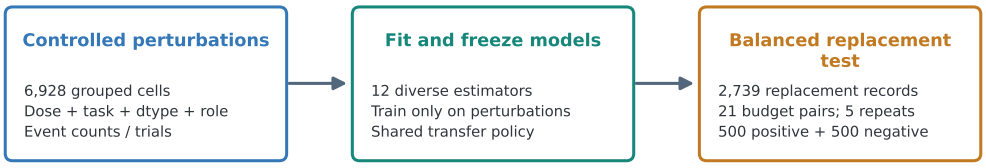


Figure 1: Models are fit on grouped controlled-perturbation observations and applied to real replacements using maximum local discrepancy. Replacement outcomes are withheld from model fitting and selection.

ison asks how model family affects average screening performance, how rankings change across risk-budget pairs, and which models retain recall on replacements with larger net accuracy drops.

2 Related Work

Numerical variation in model inference. Floating-point behavior depends on arithmetic precision, evaluation order, and accumulated rounding error [1]. Accuracy-aware compilation and hardware emulation study how numerical deviations can be bounded or exposed before deployment [2, 3]. FlashAttention-3 uses low precision and asynchronous execution for speed [5], while Yuan et al. [4] show that execution settings can alter language-model outputs. The supplied Kernel-Correctness study connects controlled perturbations to measured replacement outcomes and specifies the role taxonomy and balanced evaluation protocol used here [8].

Kernel generation and local correctness. KernelBench evaluates generated GPU kernels for efficiency and local correctness [6]. TritonBench and MultiKernelBench extend evaluation to Triton operators and additional platforms [7, 9]. We use the local discrepancy measured in this stage as a feature for predicting task-level outcomes after replacement.

Risk estimation. Isotonic regression provides a nonparametric estimate constrained to be monotone in discrepancy [10]. The comparison also includes regression and spline models [11], support-vector regression [12], nearest-neighbor methods [13], multilayer networks [14], and tree ensembles [15, 16, 17, 18]. These methods encode different assumptions about smoothness, locality, partitions, and ensemble construction. Work on preserving sample-level correctness across model updates provides a related correctness endpoint [19]. Here, model parameters remain fixed while the kernel implementation changes.

3 Methodology

Figure 1 summarizes the perturbation-to-replacement protocol. We adopt the source study's operator-role taxonomy and balanced sampling design [8].

3.1 Task and grouped observations

Let $f_{\mathrm{ref}}(x)$ be the final output of a reference model execution on request x, and let $f_{\mathrm{alt}}(x)$ be the output after perturbing or replacing one kernel. Let $C(o, y) \in \{0, 1\}$ indicate correctness against the task target y. We define

$$Z_{\mathrm{change}} = \mathbf{1}\{f_{\mathrm{alt}}(x) \neq f_{\mathrm{ref}}(x)\}, \qquad Z_{\mathrm{flip}} = \mathbf{1}\{C(f_{\mathrm{ref}}(x), y) = 1 \wedge C(f_{\mathrm{alt}}(x), y) = 0\}. \quad (1)$$

The output comparison uses the task's discrete result: top-1 label for classification, selected answer for multiple-choice QA, parsed answer for GSM8K, extracted span for extractive QA, and decoded transcript for ASR. Negative flip uses the reference-correct-to-alternative-wrong transition.

The perturbation corpus is represented as grouped-binomial observations [20]. We define a context by its model, logical kernel, dtype, and benchmark. A cell adds one perturbation setting. For endpoint e, cell i contains $k_{i,e}$ events among n_i requests, with empirical rate $r_{i,e} = k_{i,e}/n_i$. We retain event and trial counts rather than expanding rates into pseudo-independent rows. The training snapshot has 6,928 cells and 1,299 contexts spanning 13 models, seven task scopes, and FP32, FP16, and BF16.

3.2 Inputs and discrepancy mapping

The shared feature set uses four inputs: $\log_{10}(d)$, task scope, dtype, and one of 13 operator-role categories. The role taxonomy separates attention linear, attention arithmetic, attention PV, attention softmax, normalization, activation, convolution, pooling, MLP projection, generic projection, output head, logit scale, and residual scale. Model identity, candidate identity, and benchmark identity remain available for grouping and diagnostics but are not predictor features.

For a perturbation cell, d is the upper endpoint of its applied magnitude range, denoted `range_high`. For a real replacement, we use the maximum absolute difference between matched reference and optimized kernel outputs, denoted `max_abs_diff`. This max-to-max mapping preserves the upper-bound interpretation of the perturbation dose. An exact-zero replacement discrepancy receives predicted risk zero. A nonzero value outside role-specific support is evaluated at the nearest supported boundary, with fallback from task–dtype–role to task scope and then to pooled training support.

3.3 Compared models

We compare twelve models with fixed settings. The prevalence baseline predicts request-weighted training prevalence. Dose-only isotonic regression uses weighted pooled adjacent violators and smoothed rate $(s + 0.5)/(n + 1)$. Ridge regression uses $\alpha = 1$. Softmax spline logistic regression uses a spline basis of log-dose with task, dtype, and role interactions. RBF support-vector regression uses $C = 1$ and $\epsilon = 0.02$. Weighted KNN averages the 25 closest cosine-distance observations, weighted by trial count and inverse distance. The grouped MLP has two hidden layers (64 and 32 units) and minimizes trial-weighted grouped categorical cross-entropy for the nested outcomes. A decision tree is restricted to depth 2. ExtraTrees uses 120 randomized trees, depth 8, and leaf size 6. Histogram gradient boosting uses 180 iterations and at most 15 leaves. CatBoost uses CPU MultiRMSE with 220 iterations, depth 4, learning rate 0.05, and L2 regularization 5.

The multi-output random forest contains 120 trees with maximum depth 8 and minimum leaf size 6. Each split considers every available feature, and the random seed is 20260905. The forest jointly regresses the output-change and negative-flip rates. During fitting, we scale the negative-flip target by two and restore it to its original scale at prediction. A shared post-fit projection enforces $p_{\text{flip}} \leq p_{\text{change}}$.

All learned predictions are clipped to $[0, 1]$ and projected to satisfy $p_{\text{flip}} \leq p_{\text{change}}$. To apply one consistent transfer policy, each context is evaluated on a 61-point geometric dose grid. We replace raw predictions by their nondecreasing upper envelope and use an upper-step lookup for the observed replacement discrepancy. This post-fit policy makes the monotonicity and support behavior explicit for every model.

3.4 Balanced replacement evaluation

The replacement pool contains 2,739 unique records from the current packaged snapshot. We fit each model on all perturbation cells, save predictions from replacement features, and then score against replacement event counts. A risk-budget pair is $(b_{\text{change}}, b_{\text{flip}})$, whose components set the maximum acceptable rates for the two endpoints. A model flags a replacement if either estimated rate exceeds its budget. The joint-positive label uses the same rule with measured rates. We evaluate 21 pairs formed from 0.1%, 0.2%, 0.5%, 1%, 2%, and 5%, retaining pairs where the negative-flip budget is no larger than the output-change budget.

Table 1: Balanced replacement screening across twelve risk models. F1 is the mean across 21 risk-budget pairs and five repeats. The reported SD is the mean within-pair sample SD across repeats. All models use the same balanced replacement draws.

Model	F1 ↑ (mean ± SD)	Precision ↑	Joint-positive recall ↑	$> 0.5\%$ recall ↑	$> 1\%$ recall ↑	Accepted violation fraction ↓
Prevalence baseline	0.4867 ± 0.0066	0.4097	0.6000	0.6388	0.6497	0.2879
Dose-only isotonic regression	0.7468 ± 0.0104	0.6712	0.8473	0.8521	0.8480	0.2118
Ridge regression	0.8075 ± 0.0113	0.7490	0.8801	0.8154	0.7105	0.1445
Softmax spline logistic regression	0.8338 ± 0.0116	0.7954	0.8784	0.8239	0.7267	0.1353
RBF support-vector regression	0.8020 ± 0.0103	0.7386	0.8827	0.8411	0.7624	0.1444
Weighted KNN	0.8484 ± 0.0112	0.8311	0.8678	0.8042	0.6951	0.1373
Grouped MLP	0.8460 ± 0.0108	0.8261	0.8686	0.7977	0.6770	0.1381
Decision tree (depth 2)	0.7897 ± 0.0119	0.7860	0.8168	0.8506	0.7905	0.1886
Multi-output random forest	0.8411 ± 0.0115	0.8220	0.8634	0.8375	0.7638	0.1419
ExtraTrees	0.8384 ± 0.0113	0.8164	0.8638	0.8274	0.7453	0.1429
Histogram gradient boosting	0.8384 ± 0.0117	0.8098	0.8710	0.8302	0.7433	0.1385
CatBoost	0.8346 ± 0.0119	0.8017	0.8727	0.8409	0.7626	0.1392

For every pair, we sample 500 positive and 500 negative records with replacement. Within each class and net-accuracy-drop band, contexts receive equal total weight. The bands are at most 0.5%, above 0.5% through 1%, and above 1%. Net accuracy drop is the correct-to-wrong count minus the wrong-to-correct count. We repeat each pair five times and reuse the draw manifest across models, for 105,000 draw positions. F1 is averaged across pairs and repeats. Its reported standard deviation is the mean of the 21 within-pair sample standard deviations across the five repeats. Overlapping draws describe resampling variability, not independent-record uncertainty. Balanced scores measure screening discrimination and do not estimate natural replacement prevalence. We inherit master seed 9 from the reference protocol, which reports selecting it retrospectively from 128 candidates [8]. Our run does not search over seeds, so these standard deviations omit uncertainty from that earlier seed selection.

For a sampled record, $y = 1$ denotes a joint-positive replacement and $\hat{y} = 1$ denotes a model flag. Precision is $TP/(TP + FP)$, joint-positive recall is $TP/(TP + FN)$, and F_1 is their harmonic mean [21]. The accepted set consists of records with $\hat{y} = 0$. Its accepted violation fraction is $FN/(FN + TN)$, the share of accepted records that are positive. We also report recall among replacements whose net accuracy drop exceeds 0.5% and 1%. All measures are computed per draw and summarized over repeats and risk-budget pairs. Each draw has equal positive and negative counts, so precision is specific to this balanced design.

4 Experiments

4.1 Experimental setup

The run uses the current normalized project snapshot and no new GPU execution. The perturbation input contains 6,928 cells from 1,299 contexts; the replacement feature table contains 2,739 records across the same context identities. Thus, the experiment tests transfer from controlled perturbations to real replacements within represented contexts, not generalization to unseen contexts. Replacement outcomes remain withheld from fitting and model selection. The implementation uses NumPy 2.1.3, pandas 2.2.3, and scikit-learn 1.6.1. Tree and CatBoost models run on CPU. Every model uses the same operator-role mapping, dose mapping, support policy, and balanced replacement draws.

4.2 Main comparison

Table 1 reports mean performance across 21 risk-budget pairs and five fixed repetitions. A replacement is joint-positive if either measured endpoint rate exceeds its corresponding budget. The two severity columns report recall among replacements whose net accuracy drop exceeds 0.5% and 1%. Accepted violation fraction is the share of accepted replacements that are joint-positive.

Weighted KNN has the highest mean F1 (0.8484), followed by the grouped MLP (0.8460) and multi-output random forest (0.8411). Their mean within-pair repeat standard deviations are 0.0112, 0.0108, and 0.0115. The gaps between their mean F1 values are smaller than these repeat-level standard deviations, so the average ordering is descriptive. Other criteria favor different models: CatBoost has the highest joint-positive recall (0.8727), softmax spline logistic regression the lowest accepted violation fraction (0.1353), and the depth-2 decision tree the highest recall in both net-accuracy-drop groups (0.8506 and 0.7905).

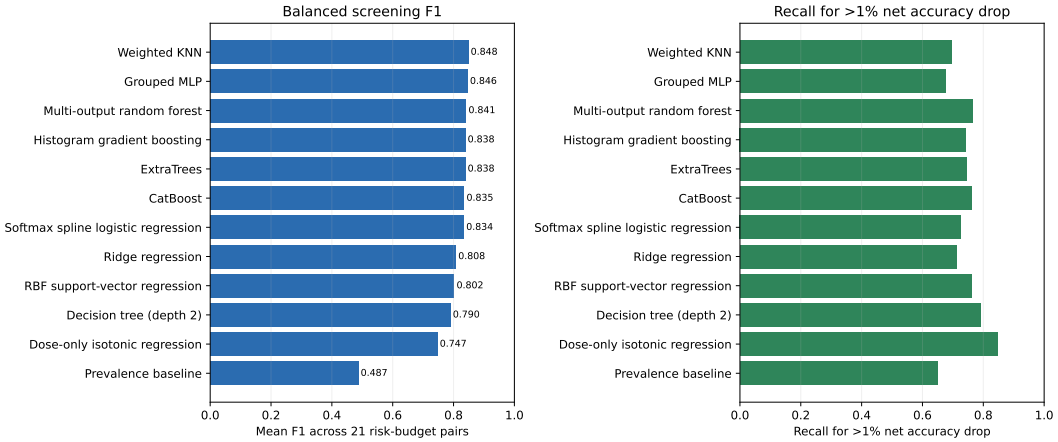


Figure 2: Mean F1 and recall among replacements with net accuracy drop above 1% across the twelve risk models. Both panels use the same balanced replacement draws.

Mean F1 rises from 0.4867 for the prevalence baseline to 0.7468 for dose-only isotonic regression and ranges from 0.7897 to 0.8484 for models that also use context features. This pattern is consistent with context features helping to distinguish replacements, but it does not isolate their effect: the models also differ in functional form, regularization, and optimization.

4.3 Interpreting balanced results

Each risk-budget pair is evaluated on draws with equal positive and negative counts. F1 and precision therefore describe balanced screening rather than performance at natural replacement prevalence. They do not estimate production precision, workload, or calibration. The shared draw manifest pairs model comparisons at the sampled-record level, but the summary reports no paired confidence intervals. Accepted violation fraction is conditioned on records the model accepts, whereas joint-positive recall is conditioned on all positive records. These metrics answer different questions and are not complements.

4.4 Sensitivity to risk-budget pairs

The mean over risk-budget pairs hides variation in the evaluation target. Table 2 summarizes each model's F1 range, standard deviation across pair-level means, and number of pairs ranked first. These quantities measure sensitivity to budget choice, whereas the five-repeat standard deviation in Table 1 measures draw-to-draw variation for a fixed pair.

Weighted KNN has the smallest across-budget standard deviation (0.0174), followed by ExtraTrees (0.0200) and the multi-output random forest (0.0229). KNN ranks first in 12 pairs and the grouped MLP in six; the grouped MLP also has the highest pair-level mean F1 (0.8957). The prevalence baseline varies most because its constant risk estimate sometimes falls below both budgets, leaving it with no positive predictions and zero F1.

4.5 Budget sensitivity and transfer support

Figure 3 shows F1 across the 21 risk-budget pairs. Model rankings change with the acceptable endpoint rates. Predictions remain frozen, so the curves reflect changes in budget-defined labels rather than refitting.

Changing the output-change budget can alter the positive labels even when the negative-flip budget stays fixed. Because each pair receives equal weight in the average, the aggregate scores may not reflect the risk tolerance of a particular deployment. A deployment decision should therefore use the relevant budget-specific confusion matrix and review cost.

The transfer policy assigns 758 records to the exact-zero state, 1,308 within role-specific dose support, 110 below support, and 563 above support. The 673 below- or above-support records receive

Table 2: F1 sensitivity across the 21 risk-budget pairs. Range endpoints and standard deviation are computed over pair-level means. First-place counts use the highest pair mean, with no ties in this run.

Model	F1 range	SD across budgets	First-place pairs
Prevalence baseline	0.000–0.765	0.3530	0
Dose-only isotonic regression	0.703–0.800	0.0293	0
Ridge regression	0.765–0.864	0.0261	0
Softmax spline logistic regression	0.790–0.874	0.0271	1
RBF support-vector regression	0.770–0.870	0.0237	1
Weighted KNN	0.817–0.871	0.0174	12
Grouped MLP	0.800–0.896	0.0263	6
Decision tree (depth 2)	0.711–0.844	0.0439	0
Multi-output random forest	0.808–0.883	0.0229	1
ExtraTrees	0.802–0.874	0.0200	0
Histogram gradient boosting	0.786–0.873	0.0237	0
CatBoost	0.776–0.876	0.0305	0

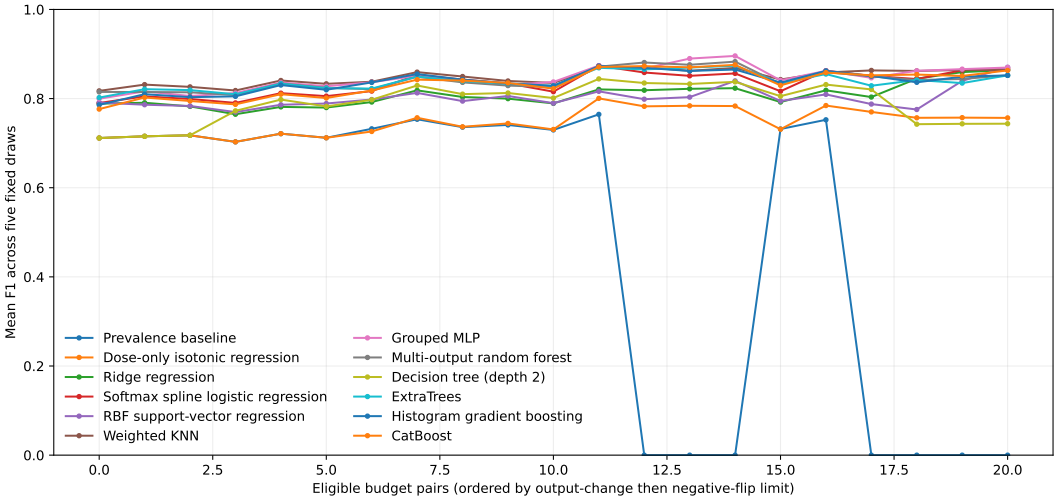


Figure 3: Mean balanced F1 across the 21 output-change/negative-flip risk-budget pairs. Predictions remain frozen while the joint-positive label changes with the budgets.

predictions through boundary clipping or fallback and therefore have weaker dose support than in-range records.

4.6 Limitations

The results cover one packaged snapshot with 13 models, seven task scopes, 1,299 contexts, and 2,739 replacements. Perturbation and replacement records share context identities, so this evaluation tests transfer between discrepancy-generating mechanisms within represented contexts, not generalization to unseen contexts. Maximum absolute discrepancy omits error sign, distribution, correlation, and accumulation across invocations. Upper-tail labels are sparse, sampling with replacement can repeat high-risk records, and boundary clipping does not guarantee calibrated estimates outside training support. The replacement labels also differ from some historical validation snapshots in the source study.

Model settings are fixed rather than selected through nested context-grouped validation, and compute budgets differ across model families. The comparison therefore measures these implementations, not the best tuned version of each family. The shared operator-role taxonomy is not evaluated through a within-model feature ablation. Balanced draws also do not estimate natural-prevalence precision, calibration, or deployment cost. The 105,000 draw positions overlap, and reported standard deviations omit uncertainty from seed selection, the replacement snapshot, and retraining. A

stronger evaluation would use an independently collected replacement pool, paired context-level intervals, and calibration measurements at the natural event rate.

5 Conclusion

We compared twelve risk models under a shared perturbation-to-replacement protocol and balanced replacement evaluation. Weighted KNN and the grouped MLP have the highest mean F1, but rankings shift across risk-budget pairs and metrics. CatBoost has the highest joint-positive recall, softmax spline logistic regression the lowest accepted violation fraction, and the shallow tree the highest recall among larger accuracy-drop cases. The leading mean F1 scores are close relative to repeat-level variation, so the evidence supports a metric- and budget-specific comparison rather than a universal ranking. These conclusions apply to contexts and discrepancy support represented in the current snapshot.

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